



DIPARTIMENTO DI **I**NGEGNERIA **E**LETTRICA E **T**ECNOLOGIE DELL'**I**NFORMAZIONE
SCUOLA POLITECNICA E DELLE SCIENZE DI BASE

LAUREA IN INGEGNERIA DELL'AUTOMAZIONE

ANALYSIS AND CONTROL OF AN HINDMARSH-ROSE NEURONAL MODEL

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Academic Year 2025/2026

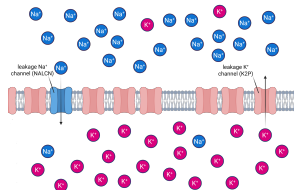
Outline

1. Context and Modeling
2. Open Loop Analysis
3. Control Strategies
4. Performance Comparison
5. Conclusions

Context: Hindmarsh-Rose model

$$\begin{cases} \dot{x}_1 = x_2 - ax_1^3 + bx_1^2 - x_3 + I + u \\ \dot{x}_2 = c - dx_1^2 - x_2 \\ \dot{x}_3 = r[s(x_1 - x_r) - x_3] \end{cases}$$

- ▶ x_1 : Membrane potential.
- ▶ x_2 : Fast Na^+/K^+ recovery current.
- ▶ x_3 : Slow Ca^{2+} -activated K^+ current.
- ▶ I, u : External stimuli and control input.



Neuronal currents

Dynamics Insight

- ▶ **Cubic term (x_1^3):** Pulls the voltage down to keep the system stable.
- ▶ **Quadratic term (x_1^2):** Pushes the voltage up to create the spike.
- ▶ **Time scale ($r \ll 1$):** Enables bursting behavior.

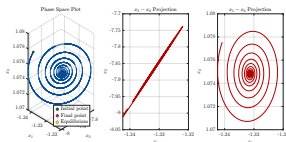
Nominal Parameters

Par.	Value
a	1.0
b	3.0
c	1.0
d	5.0
r	0.003
s	4.0
x_r	-1.6

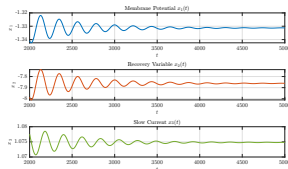
Open Loop Analysis: Bifurcations

Key Bifurcations

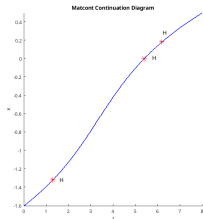
- ▶ **Subcritical Hopf:** At $I = 1.305$, focus and cycle annihilate.
- ▶ **Fold of Limit Cycles:** At $I = 1.276$, stable/unstable cycles born.
- ▶ **Transition:** From equilibrium to spiking.



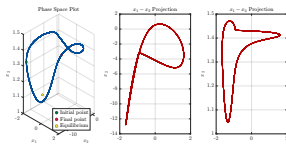
Stable Focus ($I = 1.26$)



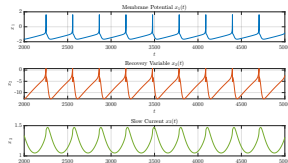
Stable Focus ($I = 1.26$)



Continuation diagram

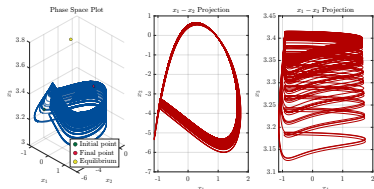


Stable Cycle ($I = 1.28$)

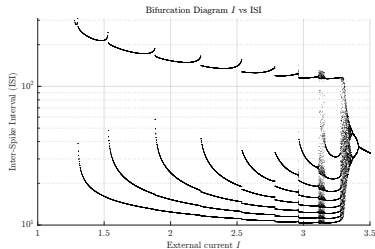


Stable Cycle ($I = 1.28$)

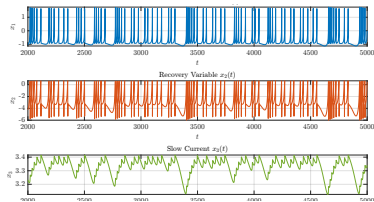
Open Loop Analysis: Chaos



Chaotic Attractor ($I = 3.31$)



Bifurcation Diagram



Chaotic Bursting ($I = 3.31$)

Chaotic Dynamics

- ▶ **Period-adding:** Increase of spikes per burst as I varies.
- ▶ **Chaos-to-chaos:** Transitions between chaotic regimes.
- ▶ **Reverse cascade:** Termination of chaos via reverse period-doubling.

Linear Control: PID

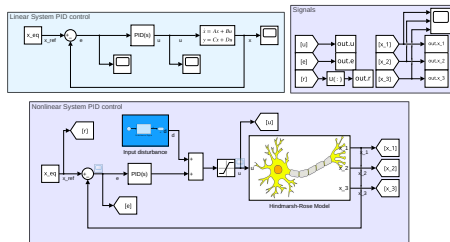
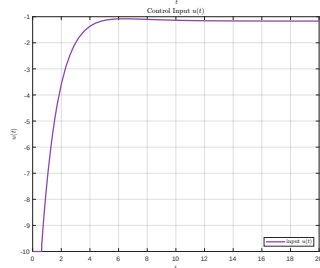
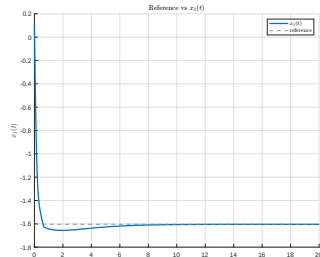


Figure: PID Simulink scheme.

Controller Characteristics

- **Tuning:** $K_P = 108.97$, $K_I = 49.48$, $K_D = 2.67$.
- **Regulation:** $T_s = 3.59t.u.$ and $S_p = 3.05\%$.
- **Limitation:** Struggles with high-frequency bursting dynamics.



PID Nominal Results

Nonlinear Control: IO FBL

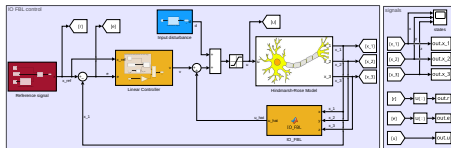
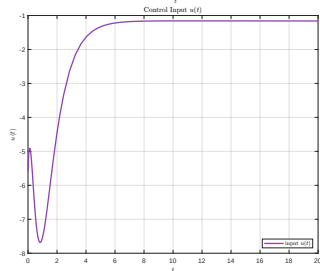
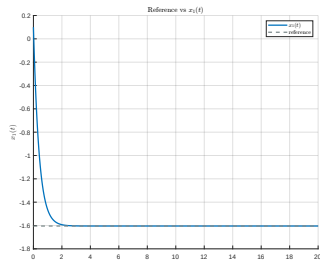


Figure: IO FBL Simulink scheme.

Feedback Linearization

- **Tuning:**

$$u = -(x_2 - ax_1^3 + bx_1^2 - x_3 + I) + K_P(x_{1,ref} - x_1),$$
 with $K_P = 2.5$
- **Regulation:** Achieves set-point regulation for the membrane potential x_1 with $T_s = 1.54t.u.$ and $S_p = 0.0\%$.
- **Limitation:** Requires an accurate analytical model; sensitive to parametric uncertainties.



FBL Nominal Results

Nonlinear Control: SMC

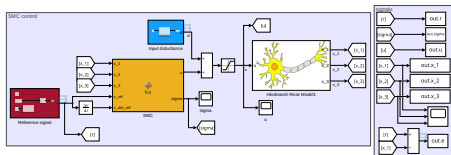
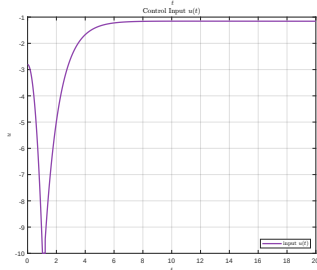
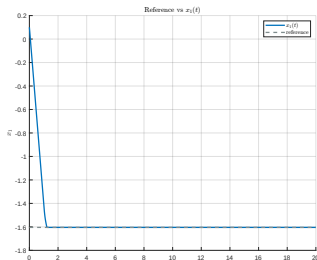


Figure: SMC Simulink scheme.

Sliding Mode Control

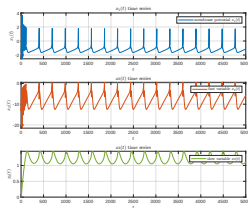
- **Tuning:**

$$u = -(x_2 - ax_1^3 + bx_1^2 - x_3 + I) - K_{smc} \text{sat}\left(\frac{\sigma}{\phi}\right),$$
 with $K_{smc} = 1.2$
- **Regulation:** Achieves set-point regulation for the membrane potential x_1 with $T_s = 1.16t.u.$ and $S_p = 0.0\%$.
- **Limitation:** Robust to parametric uncertainties and disturbances, can suffer from chattering.

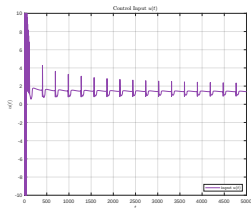


SMC Nominal Results

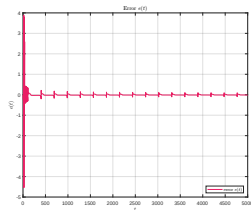
Nonlinear Control: Tracking Problem



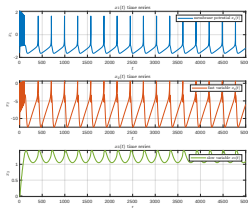
FBL control: States evolution.



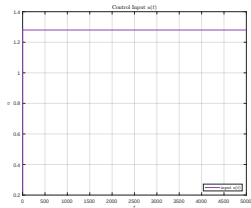
FBL control: Control input.



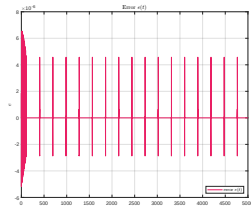
FBL control: error.



SMC control: States evolution.



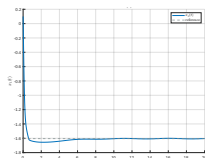
SMC control: Control input.



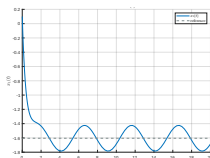
SMC control: error.

Control Strategies Comparison: Regulation

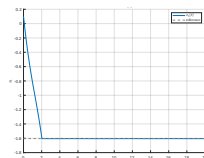
Controller	T_s (t.u.)	S_p (%)	e_∞	u_{max}	J_u
<i>Nominal Conditions</i>					
PID	3.59	3.05	2.2×10^{-6}	10.00	149.76
FBL	1.54	0.00	$< 10^{-15}$	7.68	119.63
SMC	1.16	0.00	$< 10^{-15}$	10.00	129.49
<i>Sinusoidal Input Disturbance</i>					
PID	4.07	3.04	8.10×10^{-4}	10.0	149.72
FBL	20.00	5.58	8.04×10^{-2}	10.0	172.91
SMC	1.44	0.01	1.6×10^{-7}	10.00	123.75
<i>Parametric Uncertainties</i>					
PID	4.16	3.46	2.2×10^{-6}	10.00	166.46
FBL	20.00	0.00	0.29	6.51	75.61
SMC	2.71	0.00	7.4×10^{-4}	6.73	119.91



PID response (Sinusoidal Disturbance)



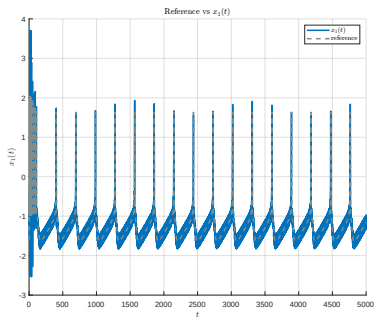
FBL response (Sinusoidal Disturbance)



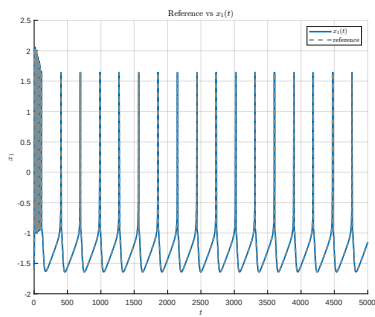
SMC response (Sinusoidal Disturbance)

Control Strategies Comparison: Tracking

Scenario	Metric	FBL	SMC
<i>Nominal Conditions</i>	RMSE	0.03	1.4×10^{-7}
	e_{max}	0.17	4×10^{-6}
<i>Input Disturbance</i>	RMSE	0.13	1.7×10^{-4}
	e_{max}	0.30	2.5×10^{-4}
<i>Parametric Unc.</i>	RMSE	0.21	4×10^{-4}
	e_{max}	0.32	6.0×10^{-4}



FBL response (Sinusoidal Disturbance)



SMC response (Sinusoidal Disturbance)

Conclusions

The Hindmarsh-Rose Model successfully captures rich neuronal behaviors, including quiescent states, periodic spiking, and chaotic bursting.

Linear Control (PID)

- ▶ Effective for regulation near equilibrium.
- ▶ Stable under disturbances, but limited by cubic nonlinearities.

Nonlinear Control (FBL)

- ▶ Superior nominal tracking ($e < 10^{-15}$).
- ▶ Lowest effort ($J_U = 119.63$).
- ▶ Highly vulnerable to parametric drift.

Nonlinear Control (SMC)

- ▶ **Most robust:** Invariant to disturbances and parametric variations.
- ▶ **Optimal speed:** Fastest settling time ($1.16t.u.$).
- ▶ Negligible error in all scenarios.

Final Verdict

SMC is the preferred strategy for high-performance neuronal tracking under realistic uncertain conditions.